

the simple bike project

start

approach

the inspiration for this project originates from the increasing challenges of accessing and repairing modern bicycles. recognising these issues, I aimed to develop a solution by creating a bike design that emphasises simplicity. the goal is to counteract the complexity of contemporary bike construction, making the process more accessible to a wider audience.

this self-build wooden bike is constructed using 26-inch wheels and beam sizes that are readily available at local hardware stores. by adopting this minimalist approach, it is ensured that the bike's essential structure remains simple and easy to assemble.

the design allows for a high degree of customisation, enabling users to adjust various measurements according to their personal preferences and body sizes. this flexibility ensures a comfortable fit and enhances the overall riding experience.

the manual provides clear, step-by-step instructions, essential measurements, and a comprehensive list of necessary components.

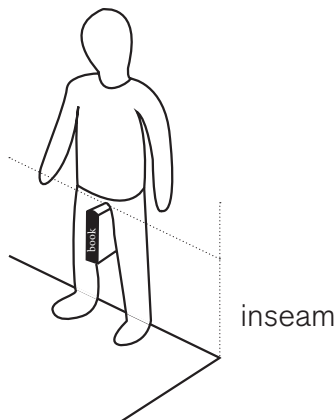
feel free to try and experiment!

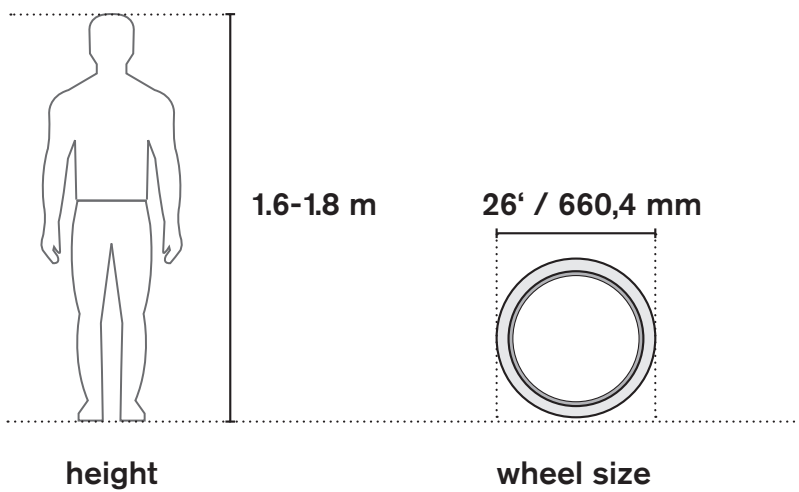


bike fit

all of the following dimensions are based on an average person with a body height between 1.6 to 1.8 meters. however, personal body measurements are key to determining the correct bike size, so it's important to *adjust*, *adapt*, and *modify* the bike to suit your individual size and preferences.

the most crucial factor in ensuring a proper bike fit is the inseam measurement—the distance from the floor (or the bottom of the foot) to the crotch. this measurement will directly influence the bike's frame size, saddle height, and overall comfort. by getting this right, you ensure that the bike feels natural to ride and helps prevent discomfort during long rides.





key terms and measurements

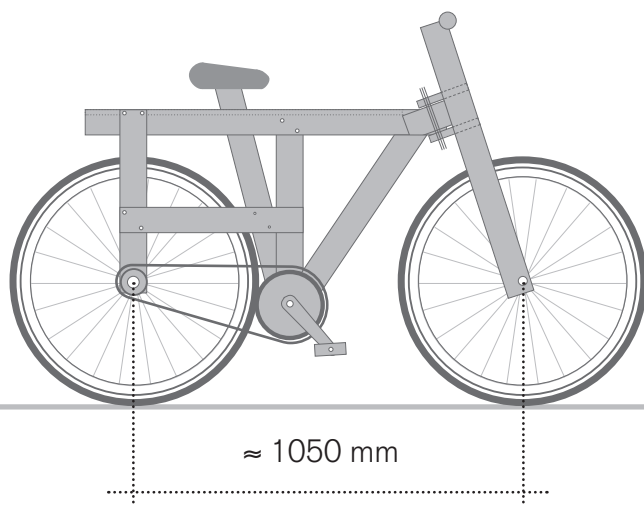
to customise and properly fit a bike there are a few factors to consider. again, this manual serves as a guideline proposing one possible solution while acknowledging that there are many other approaches.

the following measurements are suitable for this specific bike, which was the result of this project:

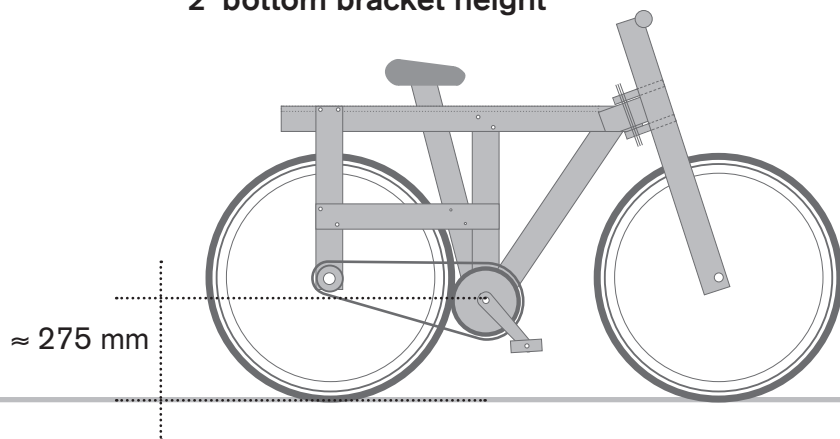
the wheelbase, the bottom bracket height, the bar height and head angle, the chainstay length, the reach, seat height and bottom bracket reach.

while there are many other measurements and angles that can affect a bike's performance, these are the most critical for building a functional and well-fitted bike.

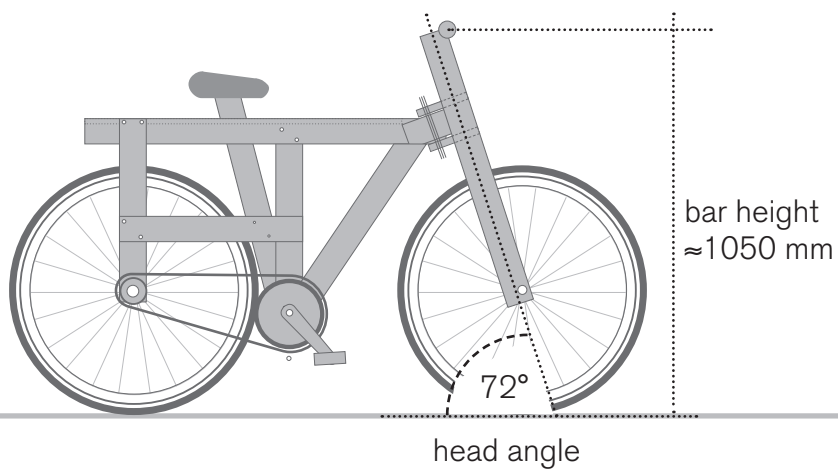
1 wheel base



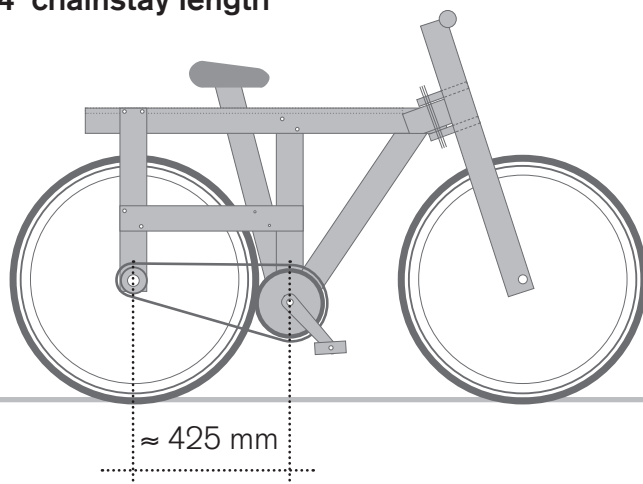
2 bottom bracket height



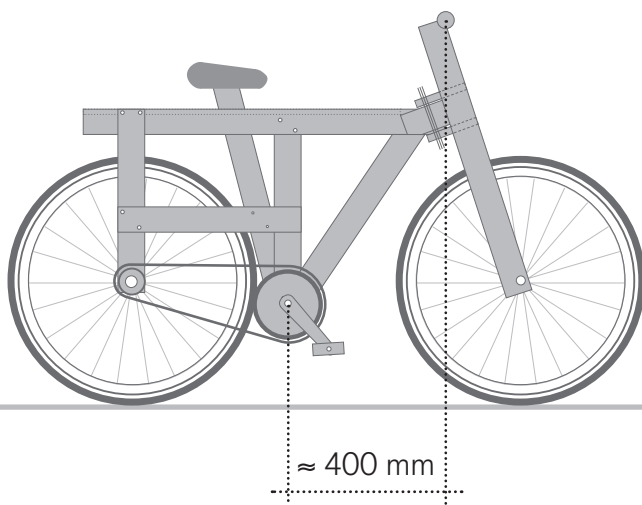
3 bar height, head angle



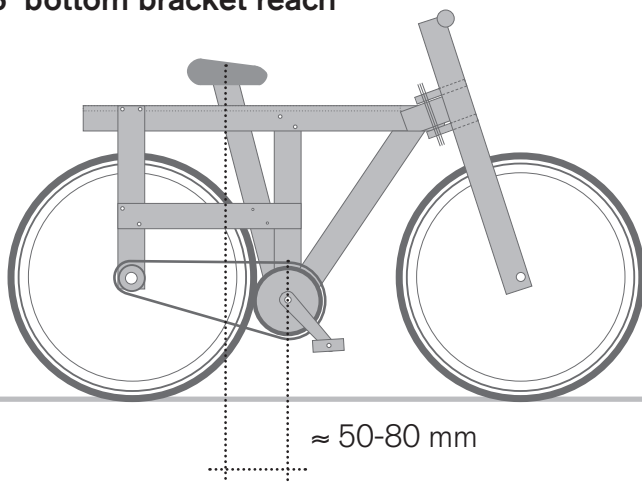
4 chainstay length



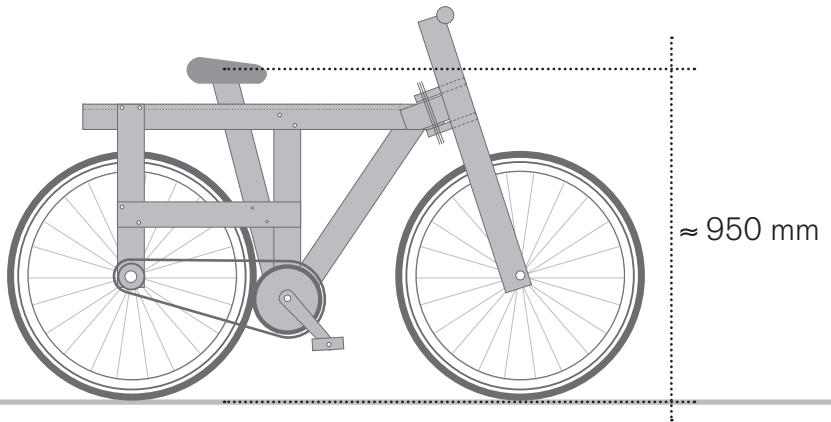
5 reach



6 bottom bracket reach



7 seat height



$$\text{inseam} \times 0.883 = \text{seat height}$$

for example:

inseam 780 mm, seat height 686.7 mm
from the center of the bottom bracket to the seat.

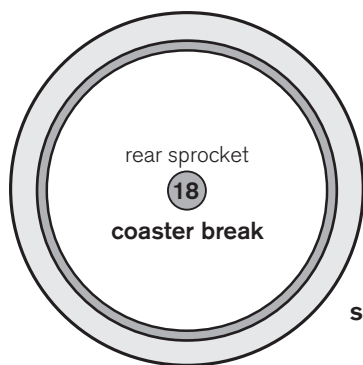
seat height floor to seat:
 $686.7 \text{ mm} + 275 \text{ mm} = 961 \text{ mm}$

bike parts

the proposed design should be considered the most basic version of a bike. it is a single-speed with a frame built around 26-inch wheels and a coaster brake, meaning it only has a pedal brake and operates with a single gear. this is achieved by using just one chainring and one rear sprocket.

one more important factor—promise it's the last one!—is the gear ratio. for single-speed bikes, the gear ratio is calculated by dividing the number of teeth on the chainring by the number of teeth on the rear sprocket. this ratio determines how easy or hard it is to pedal. for city riding, a ratio of around 2.5 to 2.7 is ideal. a higher ratio results in greater speed but requires more effort when pedaling.

26"

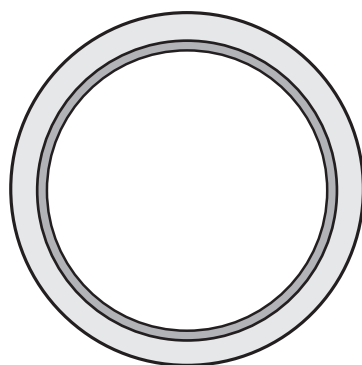


chain ring

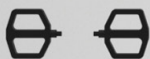
46

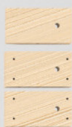
single speed

26"



ratio: $46/18 \approx 2,56$





tools

there are different tools necessary in order to be able to build this bike from scratch:

- allen key set
- wrench (10mm, 14mm, 17mm)
- drill
- drill bits: 4.5mm, 6.5mm, 10mm, 40mm
(if necessary drill aid)
- saw
(if necessary sawing aid)

components

as seen in the previous picture, those are all the parts necessary to build the bike the same way I did:

- 26-inch (590X19) rear wheel with 18T sprocket and coaster break with tube and tire
- 26-inch (590X19) front wheel with tube and tire
- pedals
- bike seat
- chain (single speed $1/2" \times 1/8"$)
- crankset 46T (170mm)
- grips
- bottom bracket (1 13mm x 68mm)
- bottom bracket shell BSA standard (68mm)

to build the frame and secure it properly, you will need screws. those are the ones I used, but again, it is totally up to you what to pick.

- 4x40 wood screws (17x)
- 4.5x50 wood screws (4x)
- 5x60 wood screws (2x)

- M6 x 60 (4x)
- M6 x 110 (7x)
- M6 x130 (2x)
- M10 x 130

- nut M10 self locking
- washer M10 (4x)
- sleeve nut M6 (11x)

this frame design is based on commonly available plank sizes. in my case I used pine wood in 21 x 69 mm, that can be found in hardware stores.
generally speaking you would need around 9.5m of wood for this bike.

the more precise measurements a drill positioning can be found on next pages.

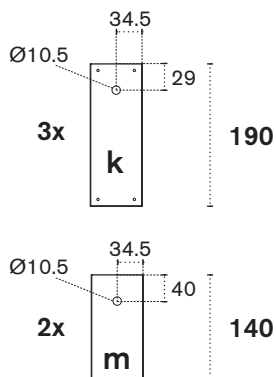
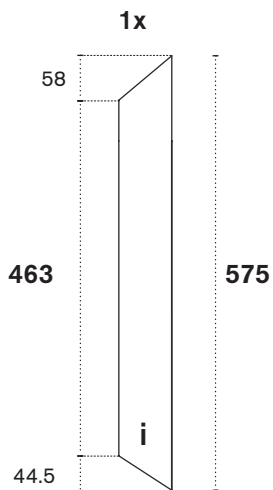
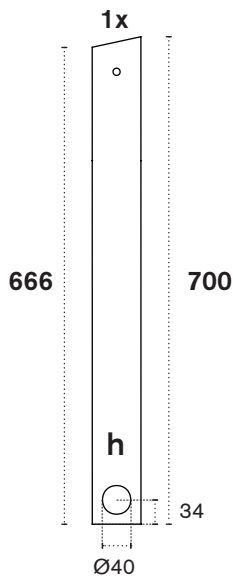
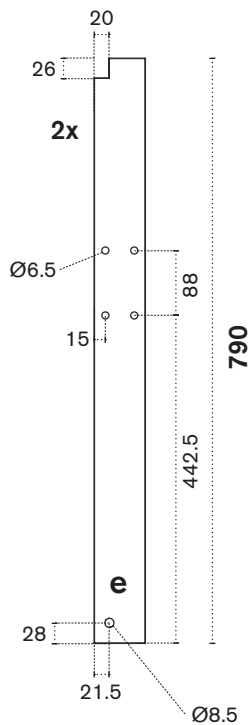
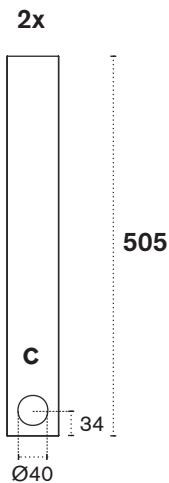
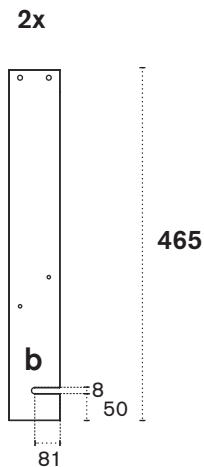
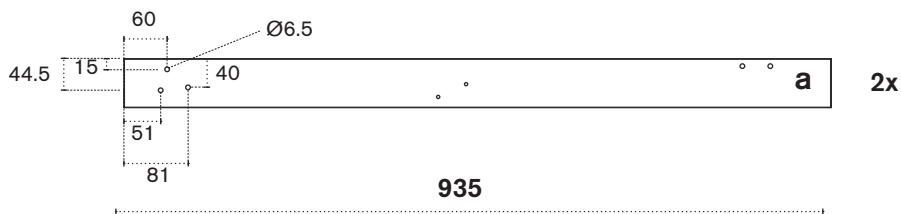
assembly

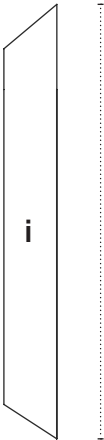
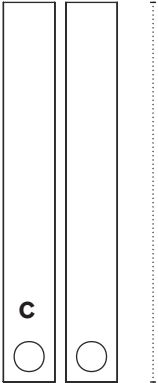
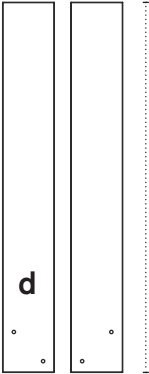
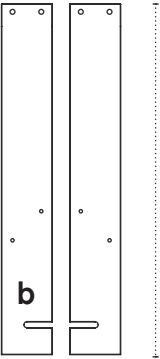
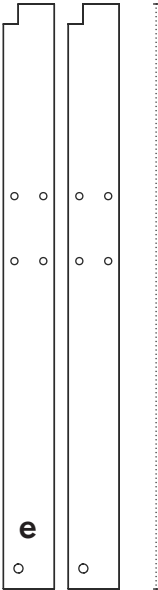
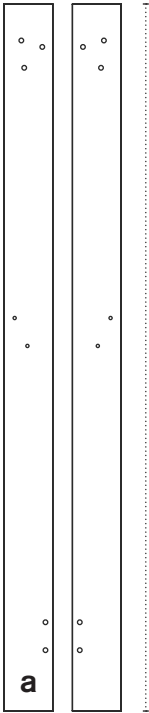
step-by-step

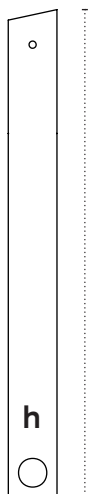
you would start by cutting all the planks to their actual size and then mark and drill the holes. it is also helpful to use the same coding as I did but again everything is totally up to you!

so as you got your planks (total around 9.5m), you should cut:

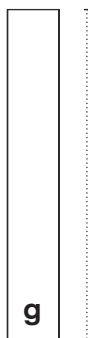
2x 935mm,
2x 790mm,
1x 700mm,
1x 575mm,
2x 505mm,
2x 494mm,
2x 465mm,
1x 441mm,
1x 341mm,
3x 190mm,
and 2x 140mm.







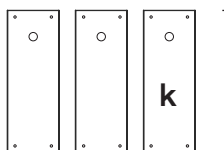
700 mm



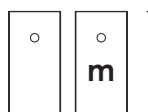
441 mm



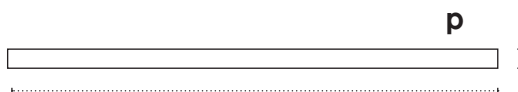
341 mm



190 mm



140 mm

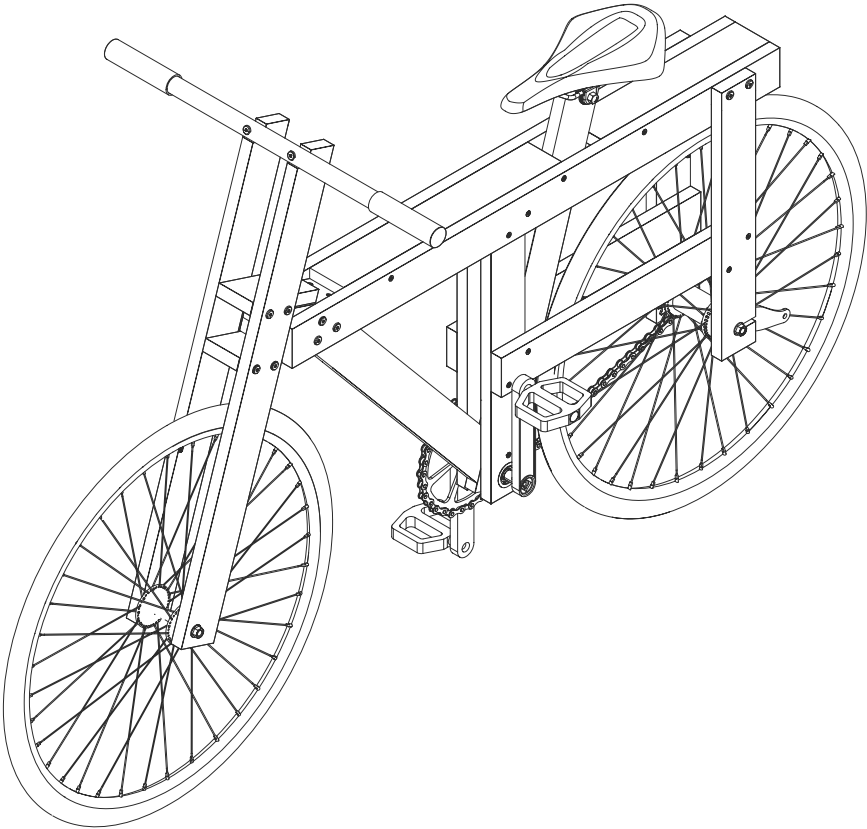


650 mm

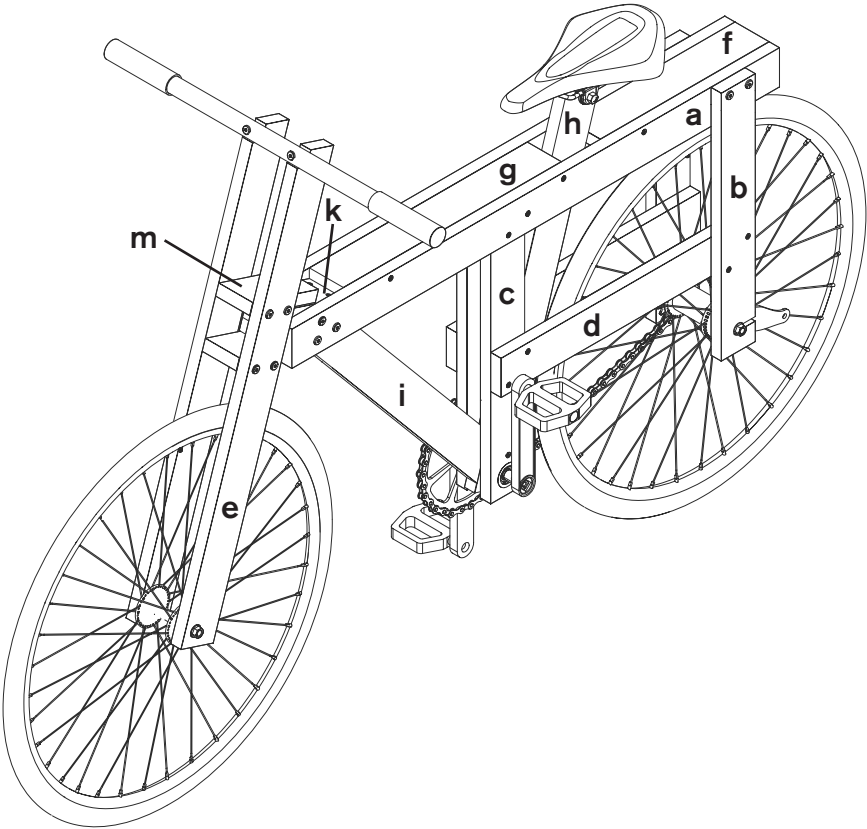
Ø27 mm

let's start assembling!

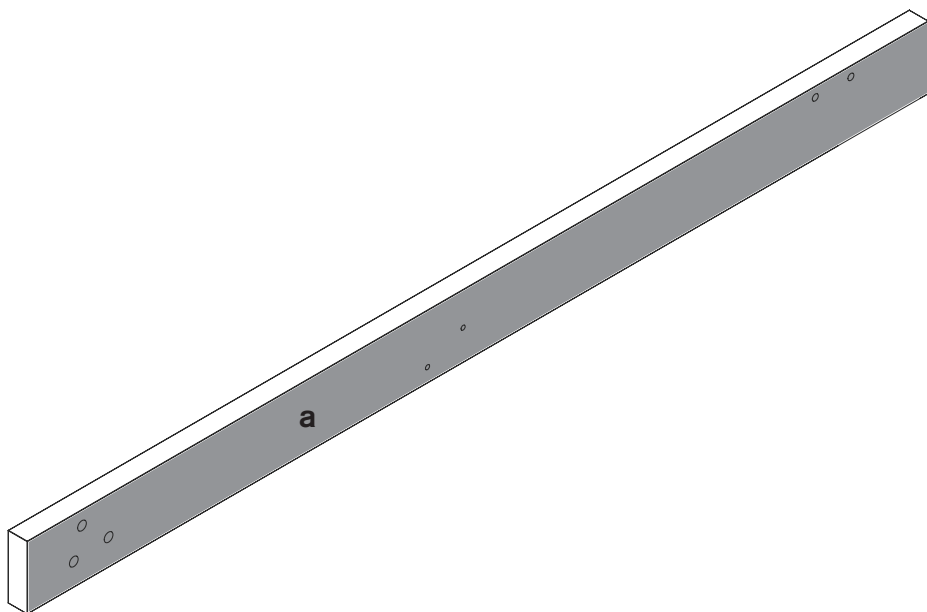
overview

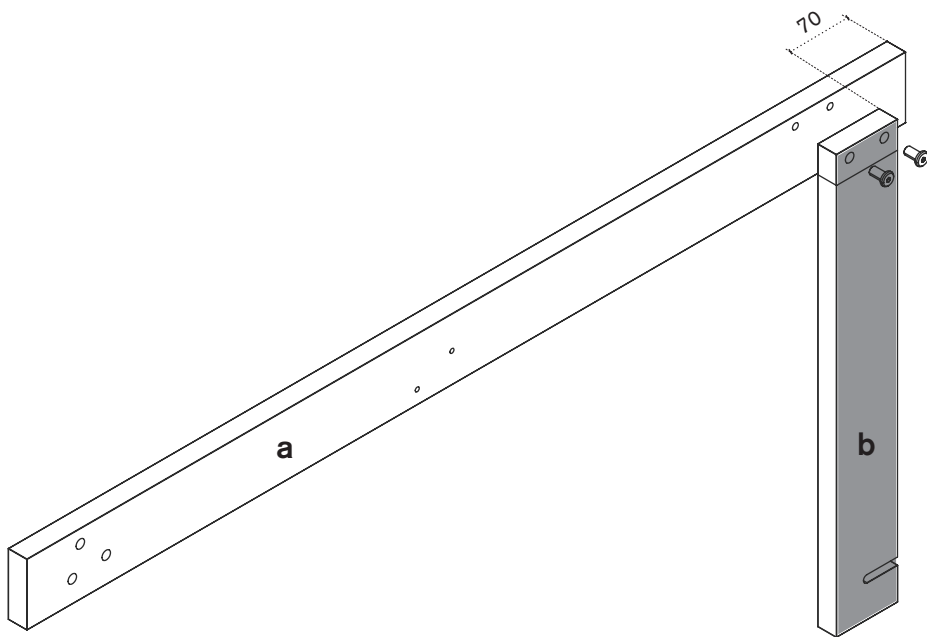


parts

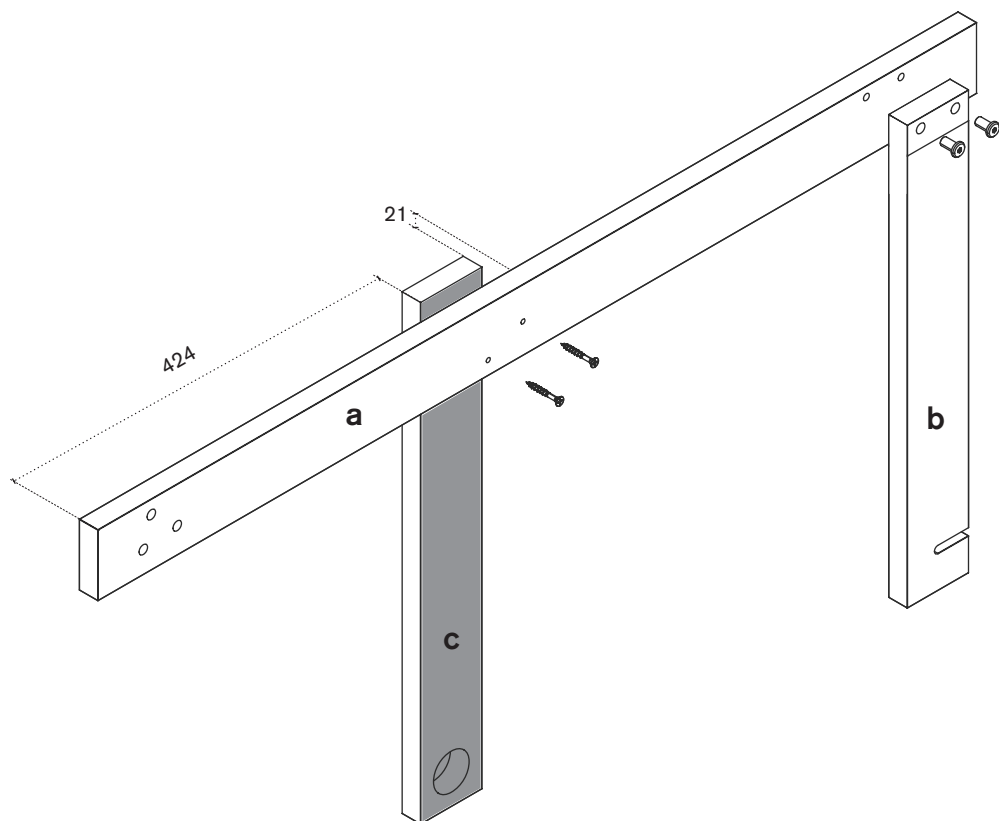


start a





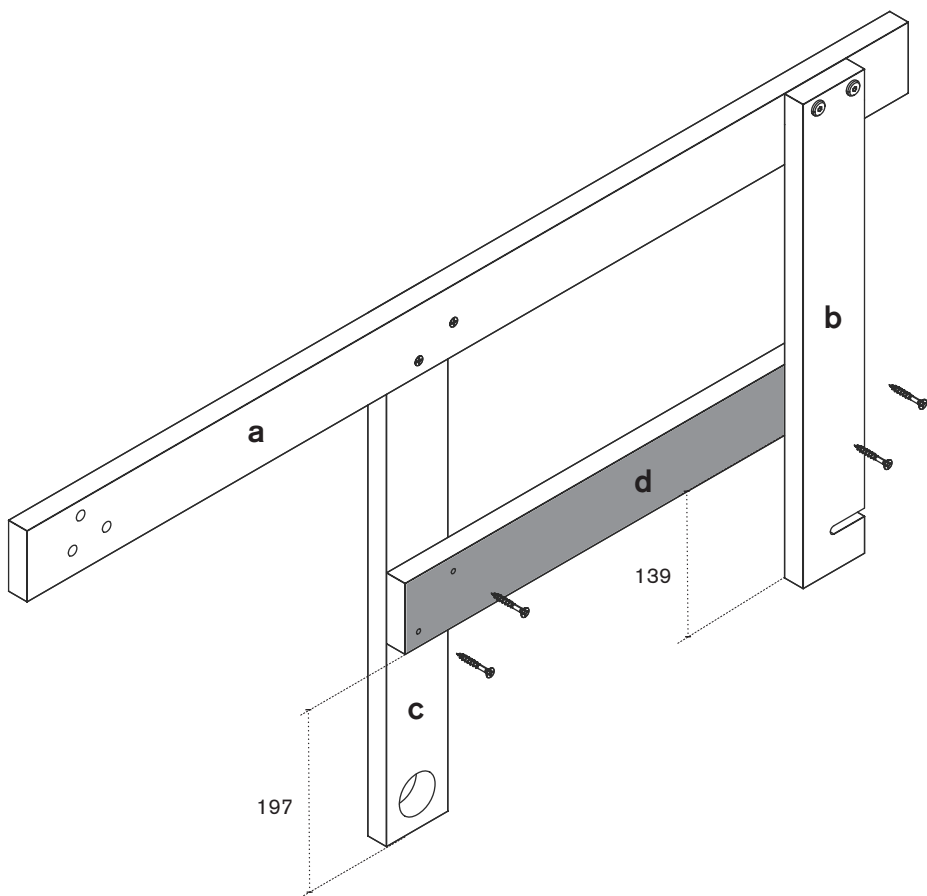
screw **b** to **a** (I put an extra wood screw in between the nuts to secure **b** in position)
distance from the rear edge: 70mm



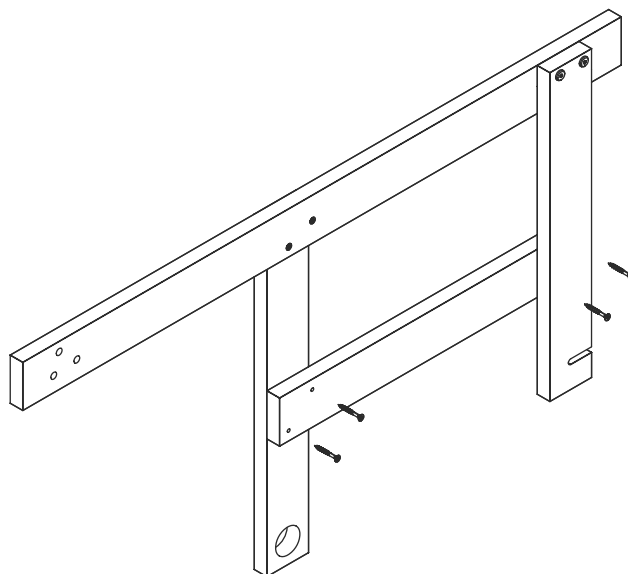
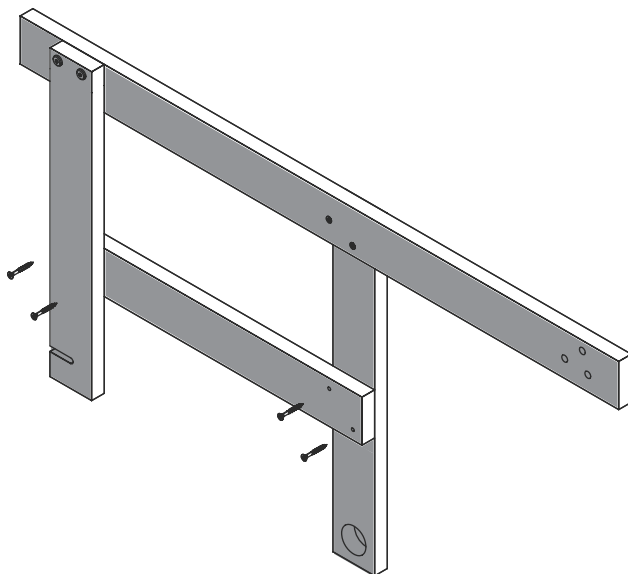
screw **c** to **a**.

distance to the front edge: 424 mm

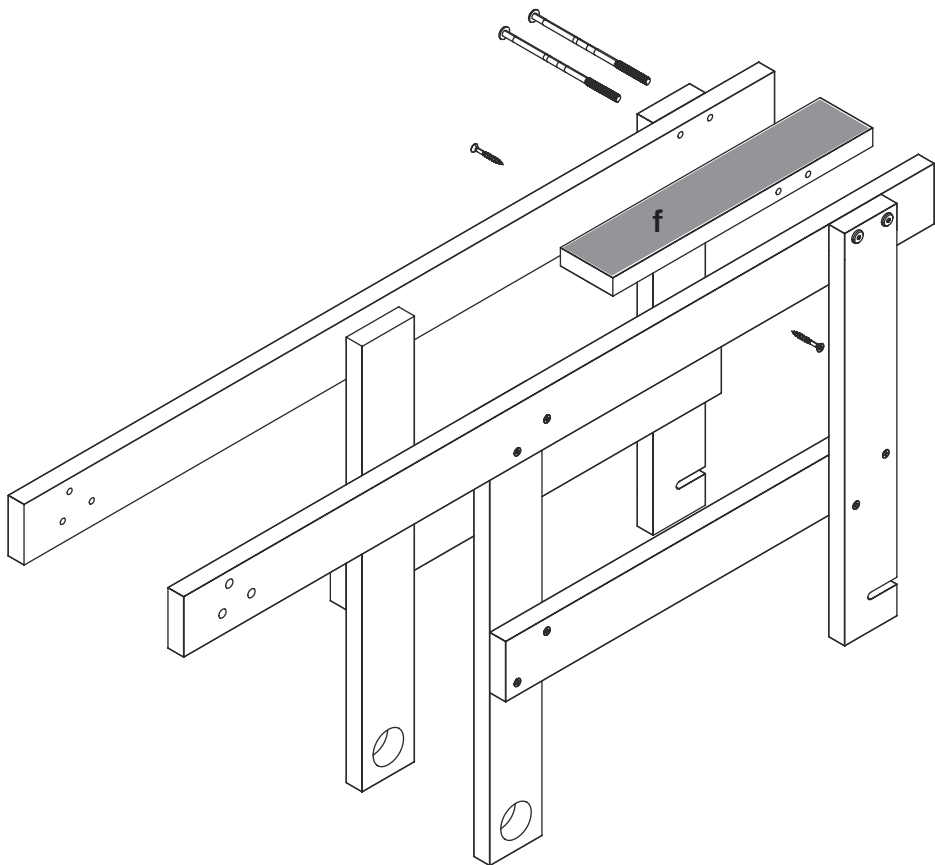
offset **c** from top edge of **a**: 21 mm



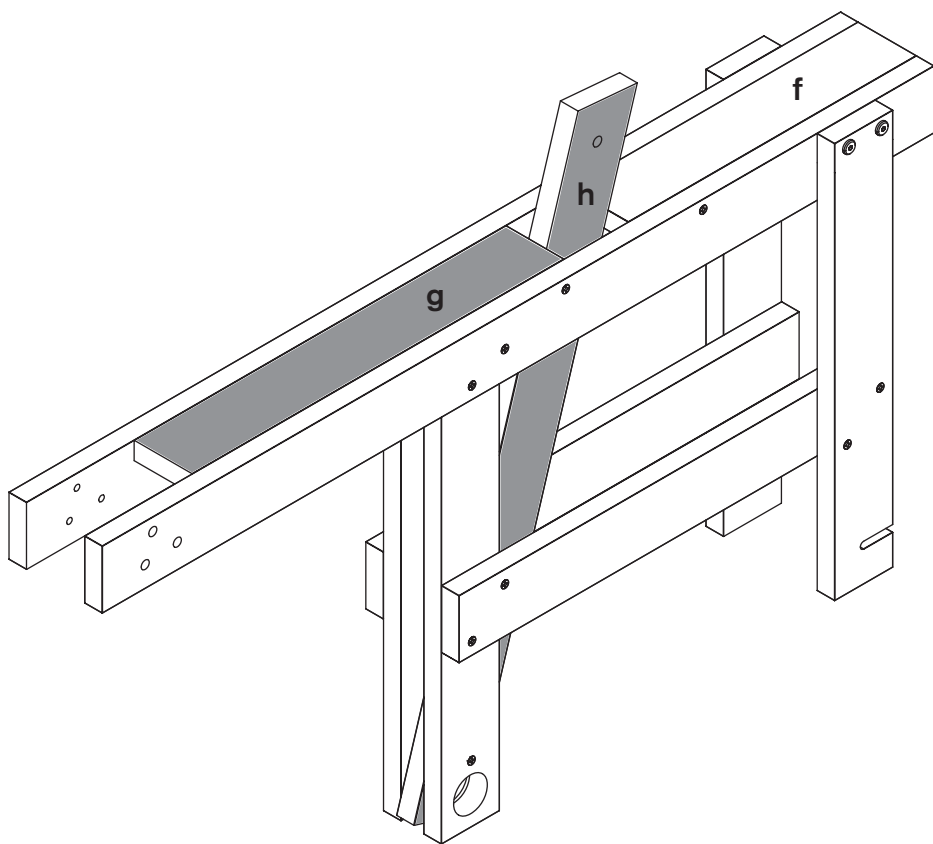
screw **d** to **b** and **c**.
distance seen in the drawing.
the first frame is now completed.



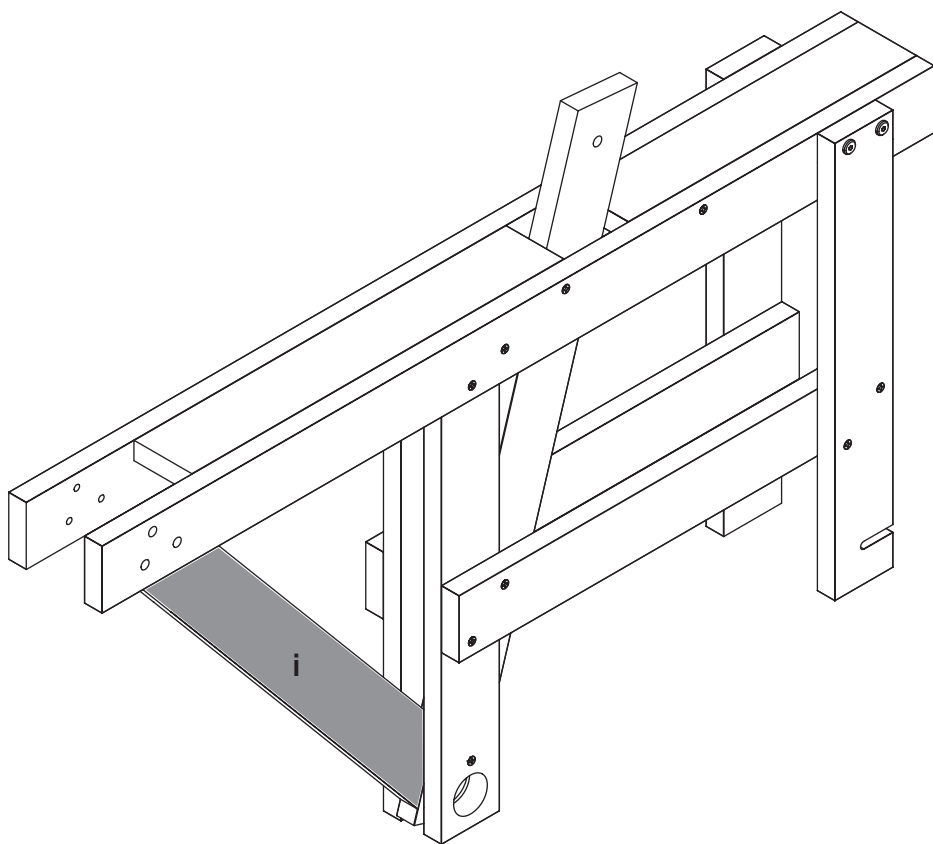
follow the same steps to build the other frame, remember that it's mirrored.



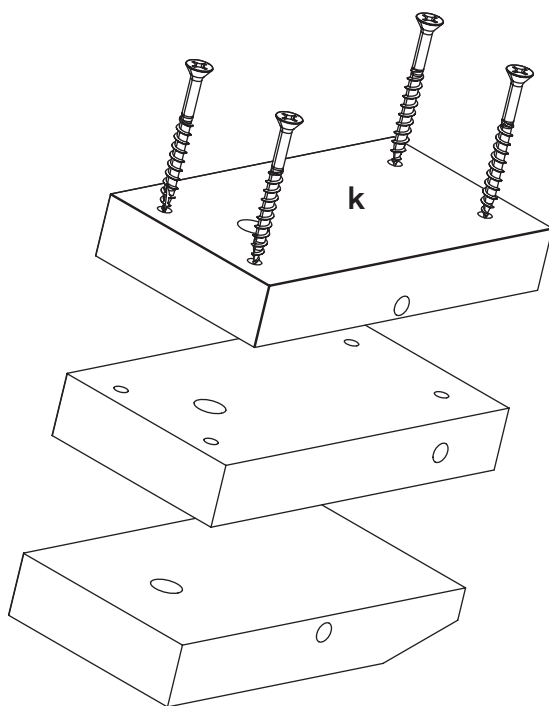
connect both frames using part **f**.



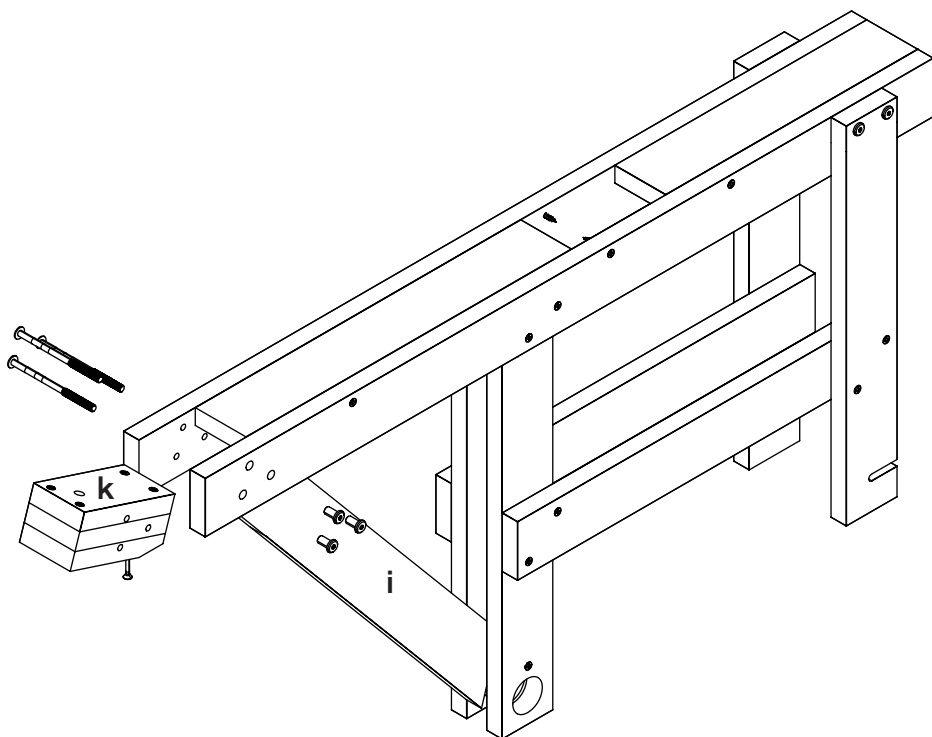
attach **g** to both frames and fit **h** into the gap between **f** and **g**, so that the hole for the bottom bracket fits.



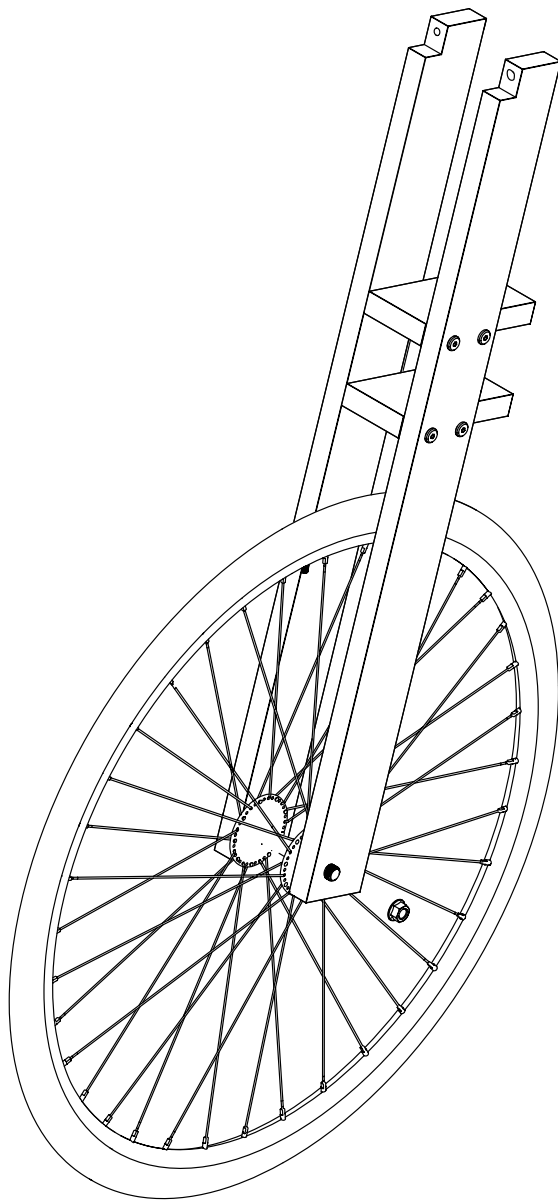
attach **i** to the frame fitting the angle of **h**.



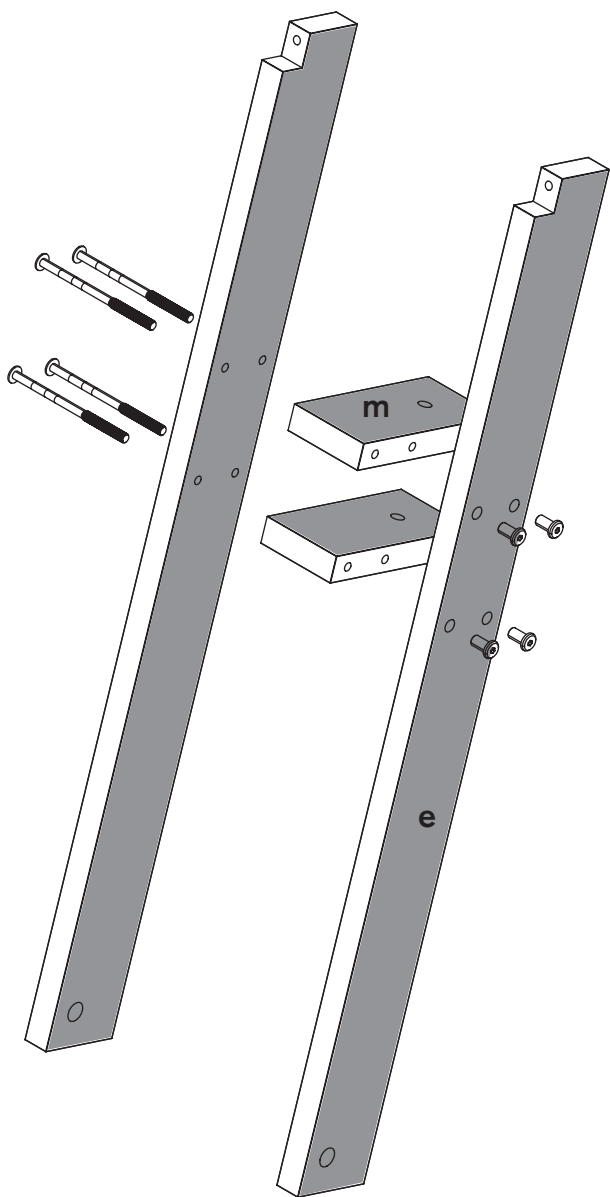
screw together the 3 pieces for the steering construction.
you could cut or sand an angle on the bottom piece of **k**.



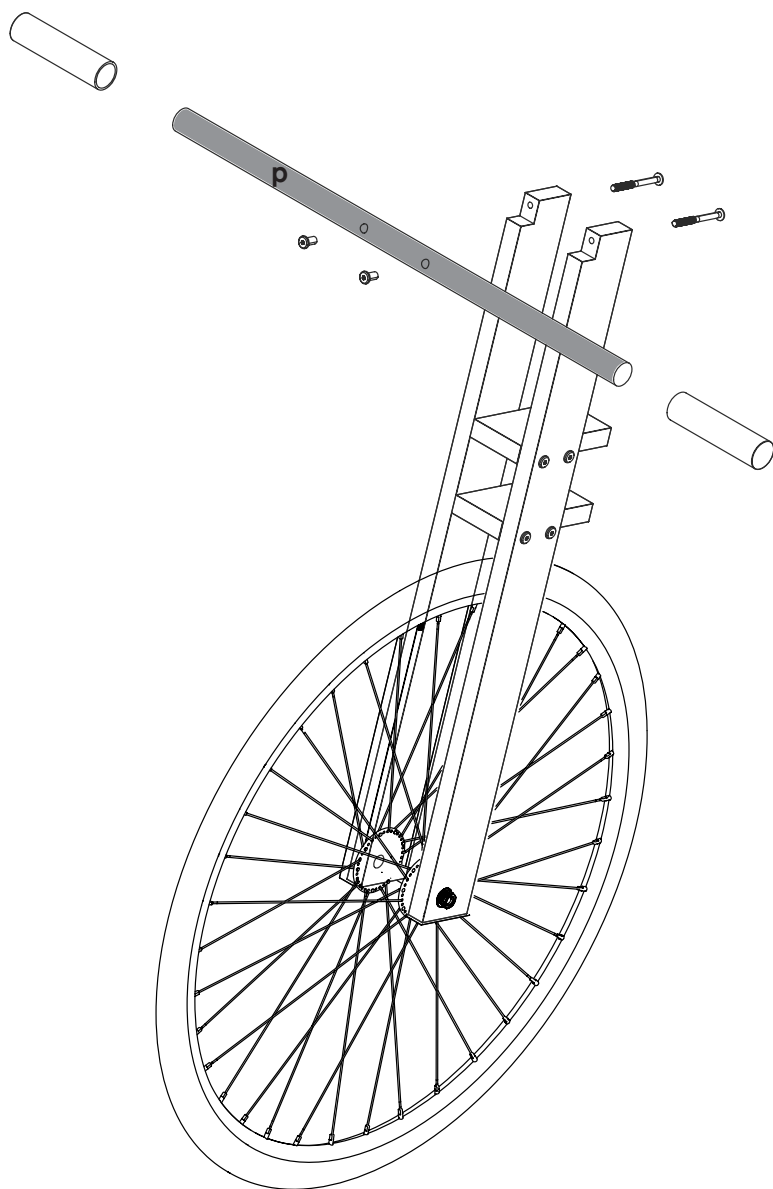
attach the steering piece to the completed frame and **i**.



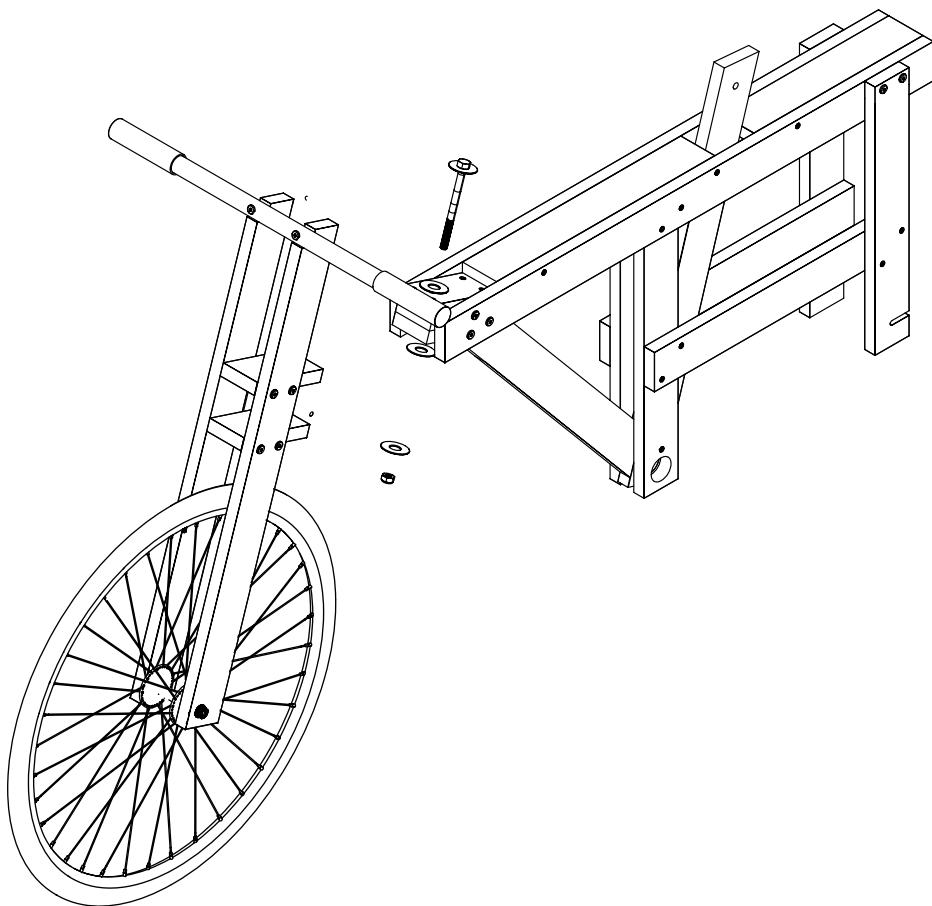
attach the front wheel to the fork, before continuing with the next step and actually assembling the fork.



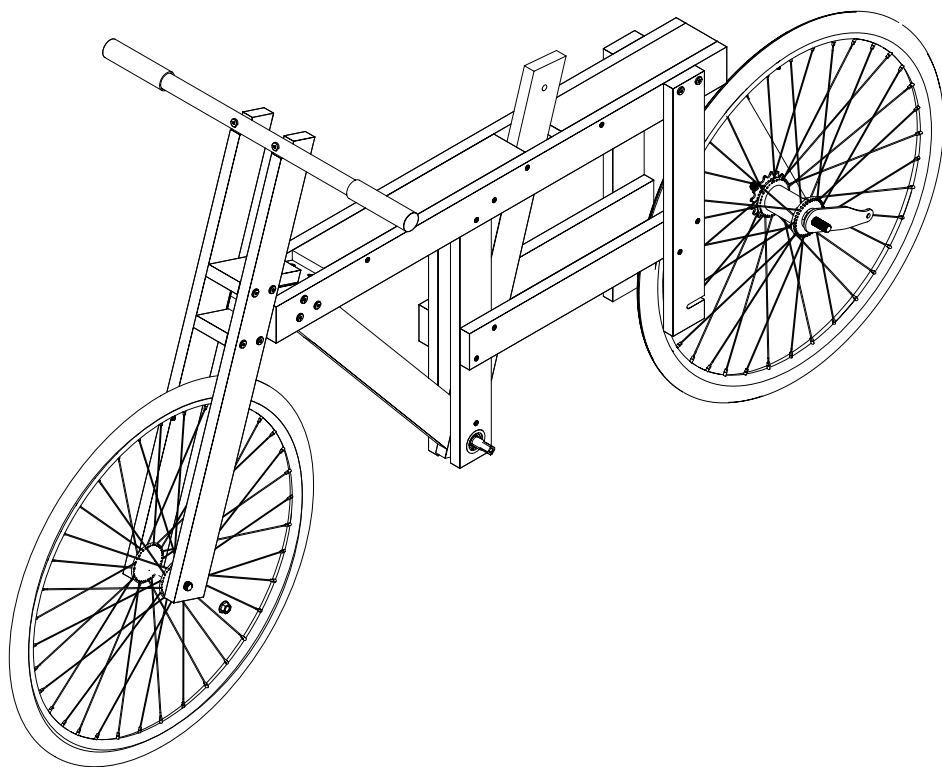
screw together both fork parts (**e**) using both pieces (**m**) for the steering construction.



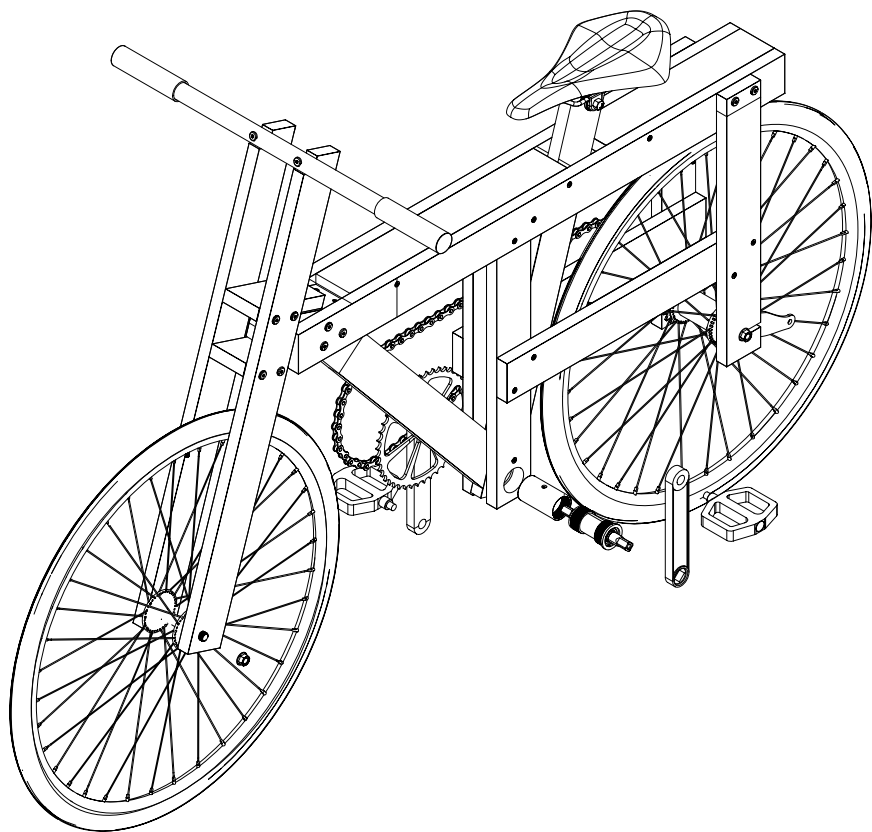
attach the handlebar **p** fitting into the cutouts in the fork and center it.



attach the fork with the front wheel to the frame using the M10 screw and put each washer wherever there would be wood on wood friction (4 in total) and fix with self locking nut on the bottom.



attach the rear wheel to the frame, but don't lock just yet.



first attach the bottom bracket shell, then fit the the bottom bracket into it (be aware of L and R), attach the chainset and pedals. now you can fit the chain and lock the rear wheel. fix the seat to your preferences.

ready to ride! (hopefully*)

*just a quick note: I can't take responsibility for the safety of the construction. it is meant to inspire you to start building and to help you tackle upcoming difficulties and maintenance with more confidence. it's up to you to ensure your bike is safe and ready to ride, but I believe in your ability to make it happen!

stay safe!